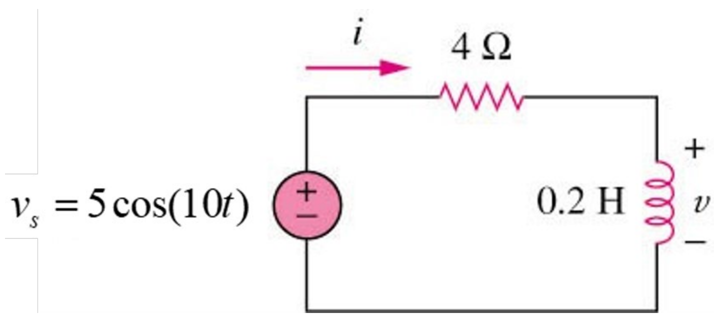
ENGR 2303 Electrical Science
Review Material for Test 4 (7/28/26 from 9-10:50 am)

The test will cover sinusoidal steady-state analysis

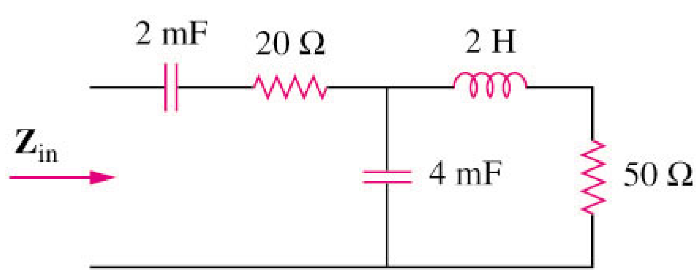
- Phasor relationships for circuit elements
- Impedance and admittance
- Kirchhoff's laws in the frequency domain
- Impedance combinations
- Nodal Analysis
- Mesh Analysis
- Superposition Theorem
- Source Transformation
- Thevenin and Norton Equivalent Circuits
- Op-Amp

Sample Questions

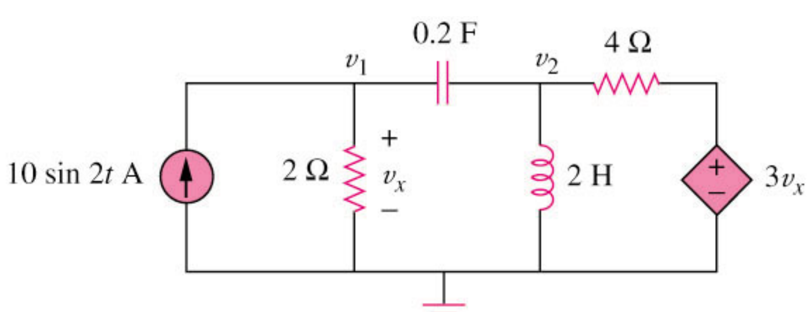
Q1) Refer to Figure below, determine *v(t)* and *i(t)*.



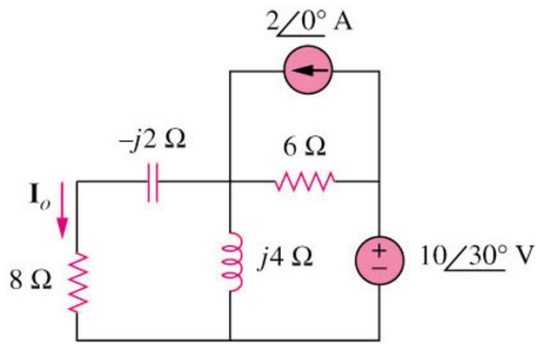
Q2) Determine the input impedance of the circuit in figure below at ω =10 rad/s.



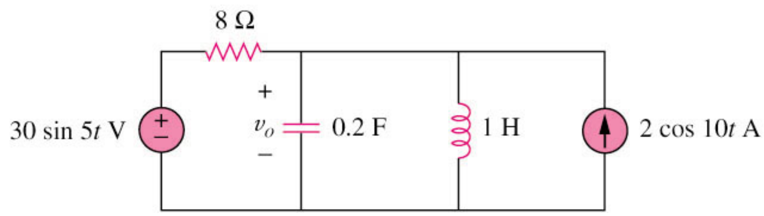
Q3) Using nodal analysis, find v_1 and v_2 in the circuit of figure below.



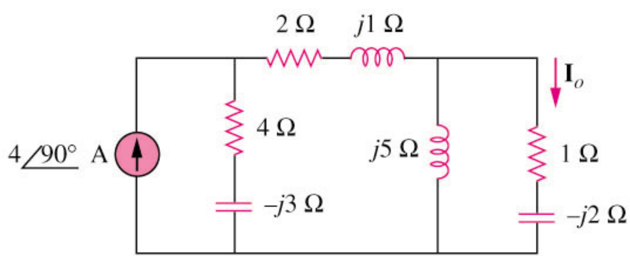
Q4) Find I_o in the following figure using mesh analysis.



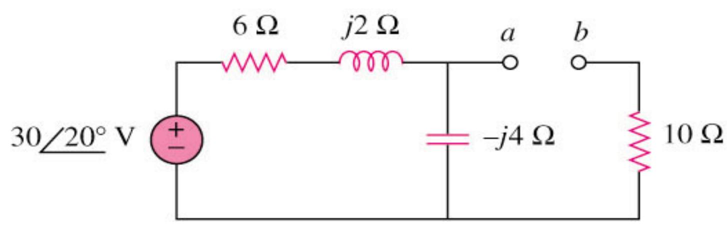
Q5) Calculate v_o in the circuit of figure shown below using the superposition theorem.



Q6) Find Io in the circuit of figure below using the concept of source transformation.



Q7) Find the Thevenin equivalent at terminals a–b of the circuit below.



Q8) Evaluate the voltage gain $\mathbf{A}_v = \mathbf{V}_o \backslash \mathbf{V}_s$ in the op amp circuit. Find $\mathbf{A}_v$ at $\omega = 0$, $\omega \to \infty$, $\omega = 1/R_1C_1$, and $\omega = 1/R_2C_2$.